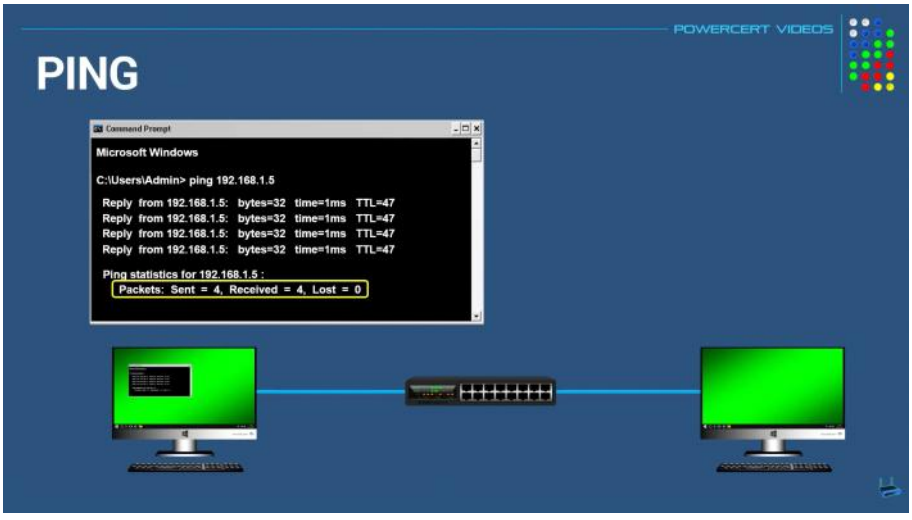
Windows Command Line Tools

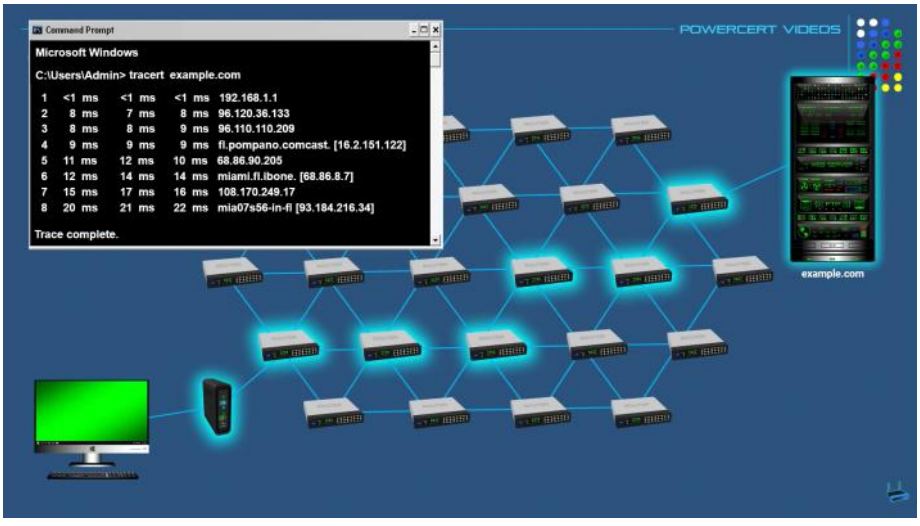
Monday, April 20, 2026 12:02 AM

Windows command line tools are powerful utilities for troubleshooting computer and network connectivity issues. Below is a summary of the key tools covered:

Ping: Checks network connectivity between your computer and a device or server by sending data packets and waiting for a response.



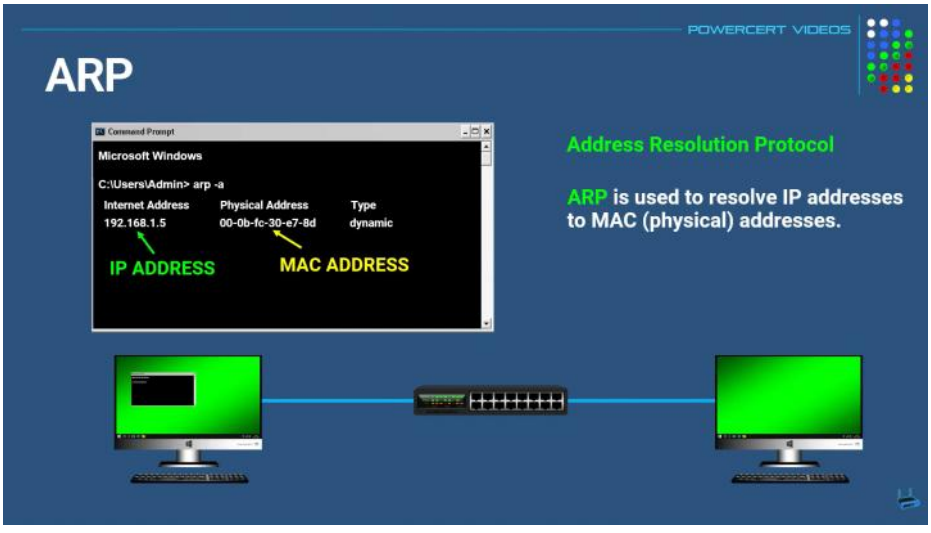
Tracert (Trace Route): Maps the path a data packet takes to its destination, identifying routers (hops) and helping pinpoint potential bottlenecks.



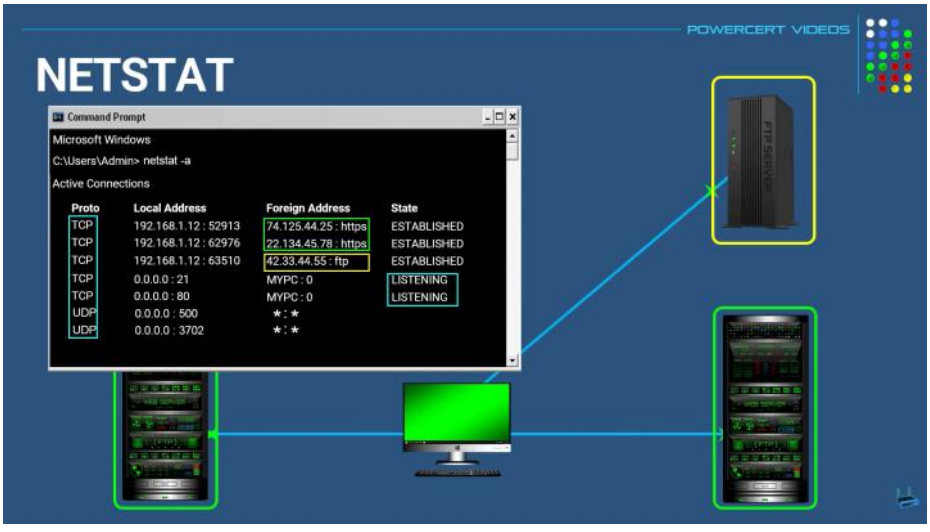
Pathping: Combines features of Ping and Tracert, providing detailed statistics on packet loss at each router along a network path.

NSlookup: Used to troubleshoot DNS issues by resolving domain names into their corresponding IP addresses.

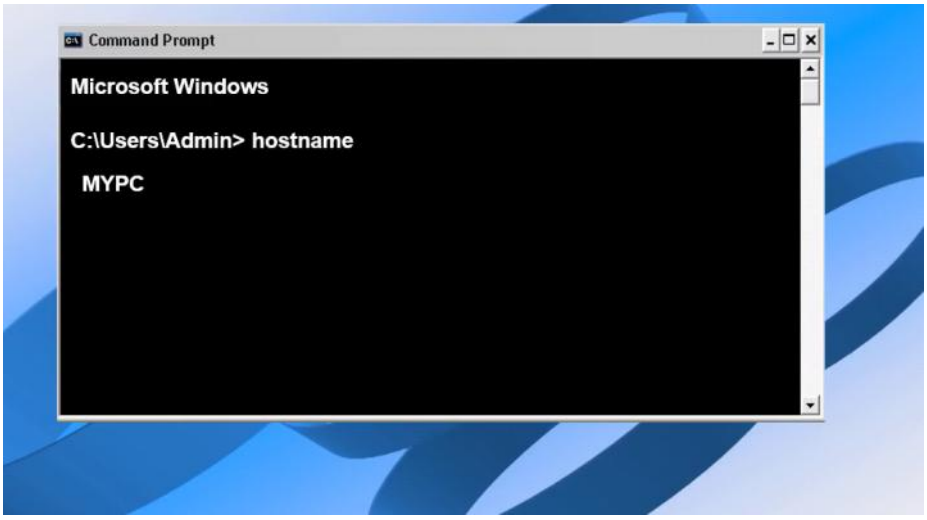
ARP: Manages and displays the Address Resolution Protocol cache, which maps IP addresses to physical MAC addresses for local network communication.



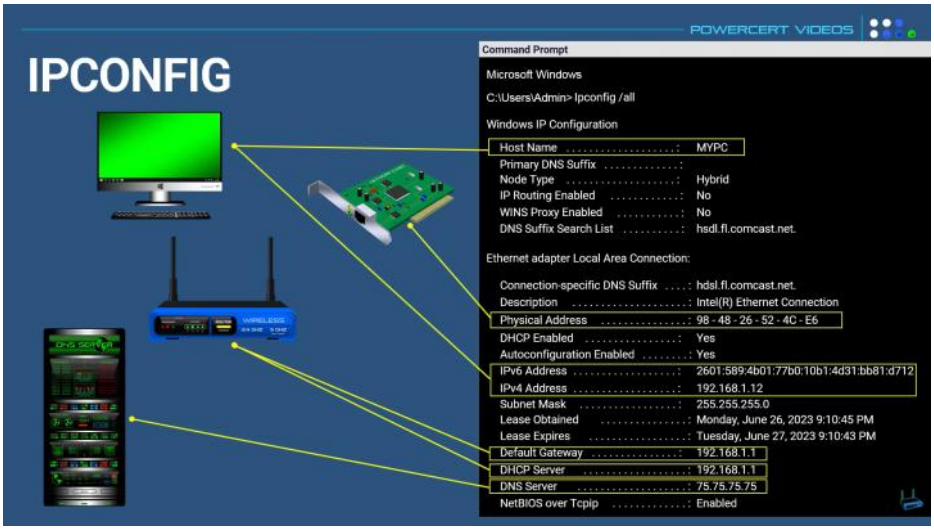
Netstat: Displays active network connections and identifies open ports, showing what the computer is currently communicating with.



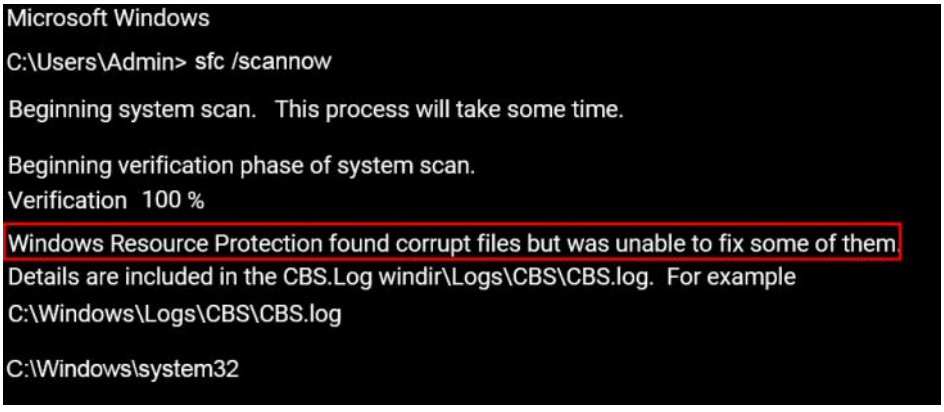
Hostname: A simple utility to display the configured name of your computer.

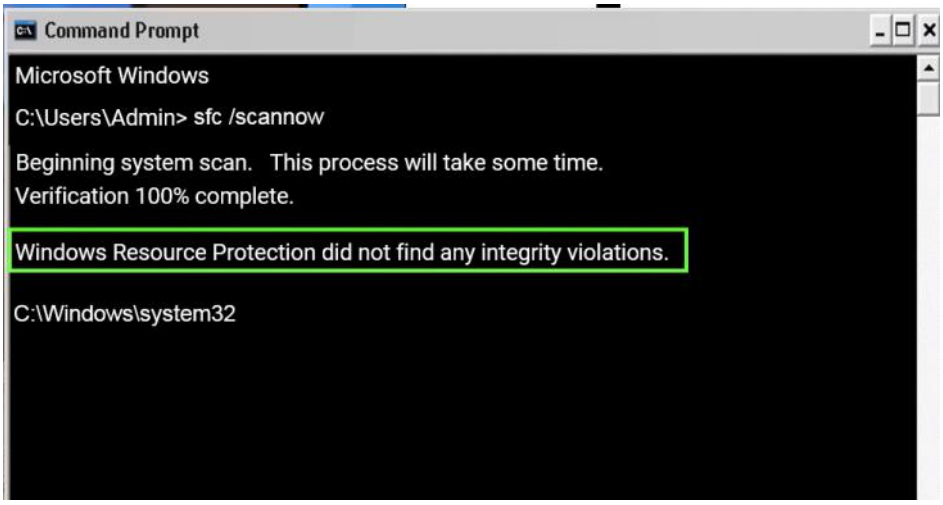


IPconfig: Shows the current TCP/IP network configuration, including IP address, default gateway, and DNS settings.

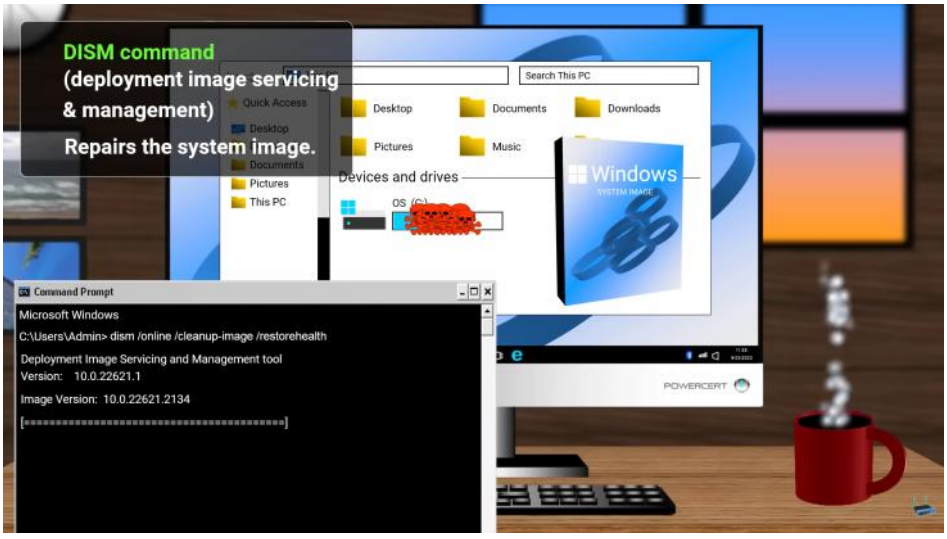
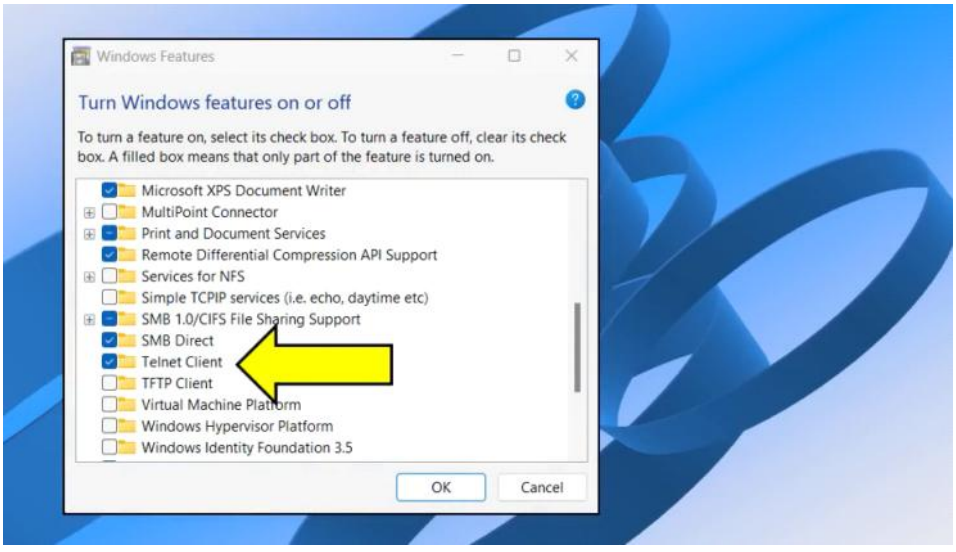


SFC (System File Checker): Scans for and repairs corrupted Windows operating system files using a system image.





DISM: Repairs the Windows system image itself, which is often required if SFC cannot successfully restore corrupted files.



Telnet: A terminal emulation program used to access and manage remote servers or network devices; note that it sends data in clear text, making it less secure for sensitive tasks.